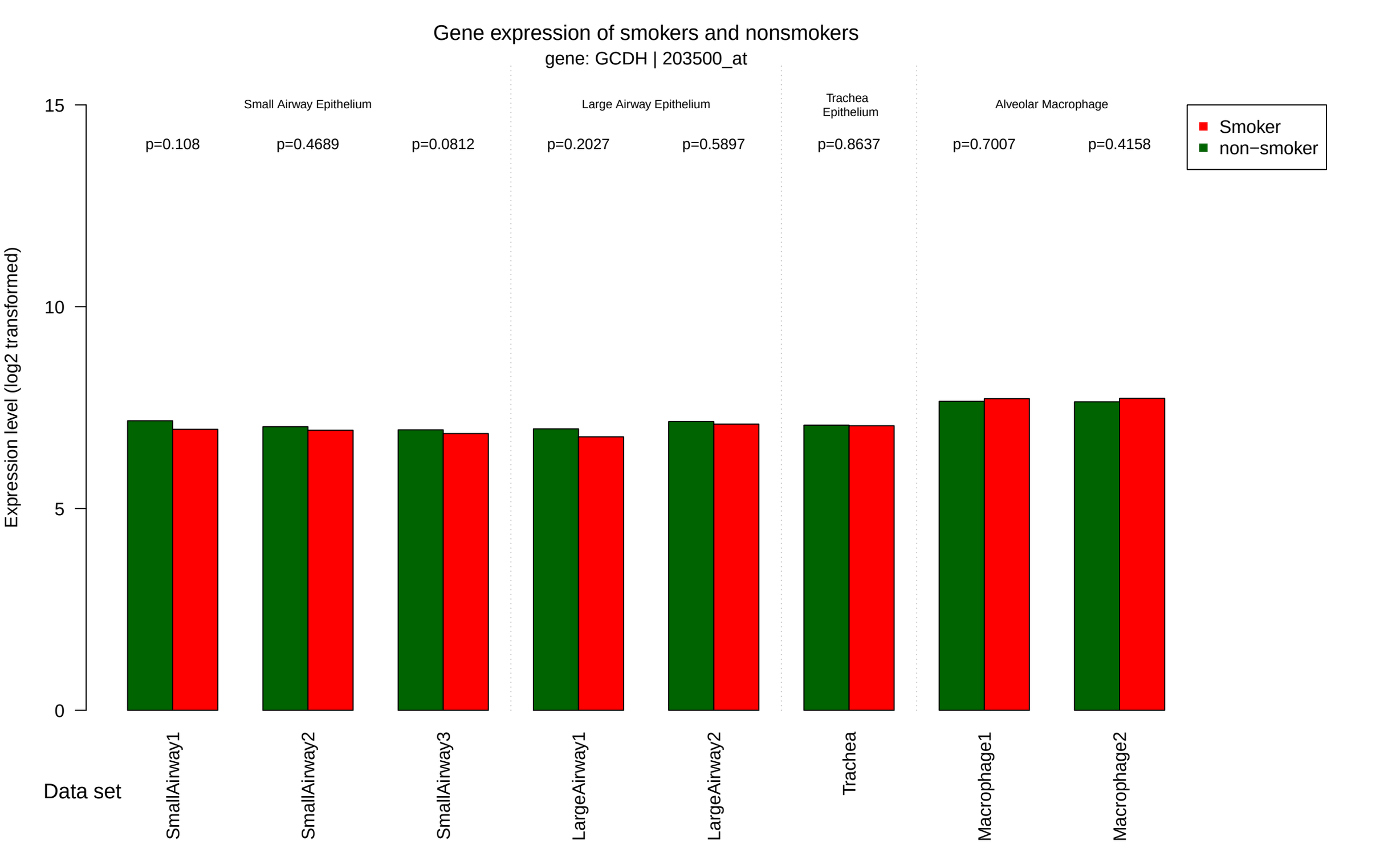
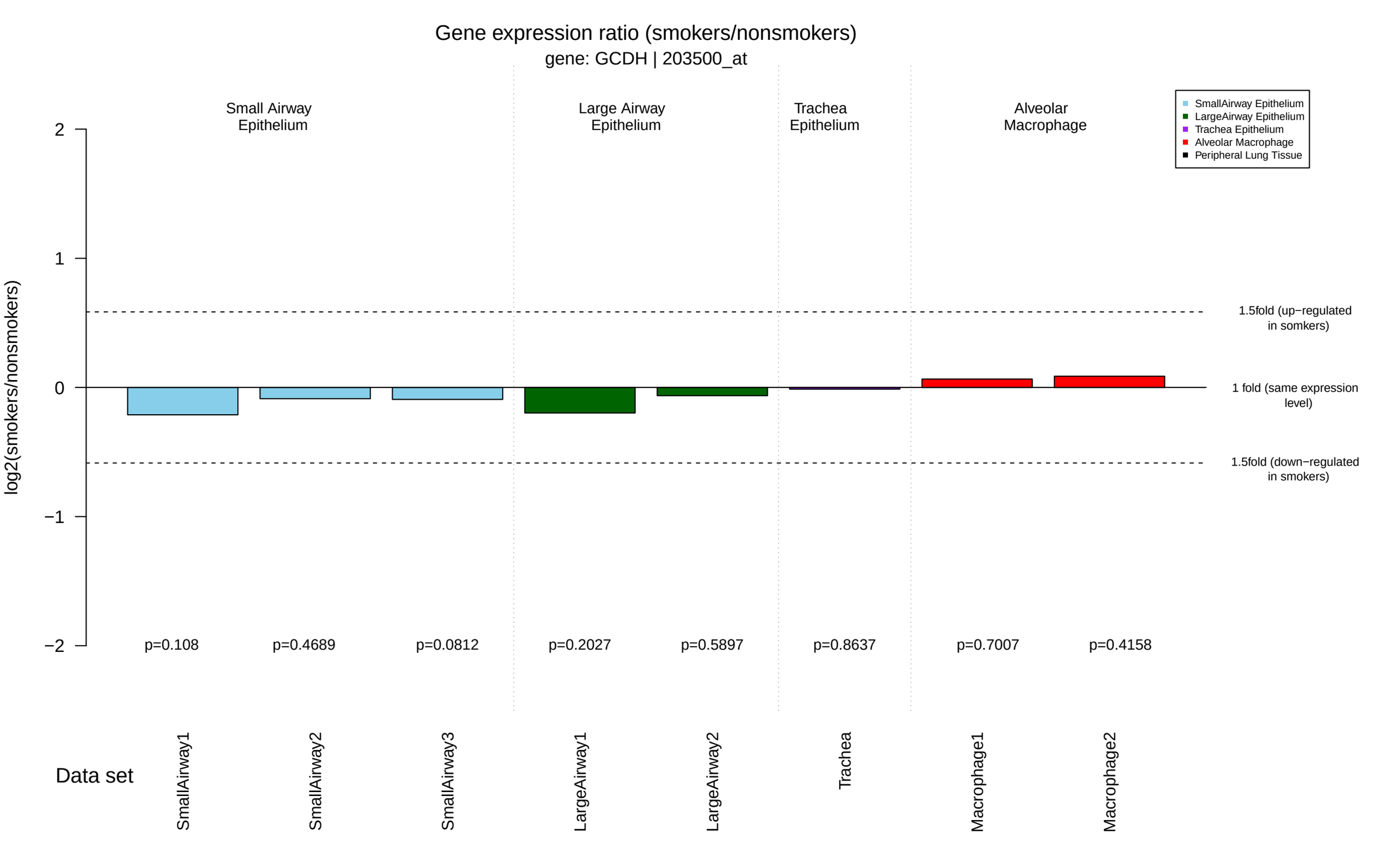
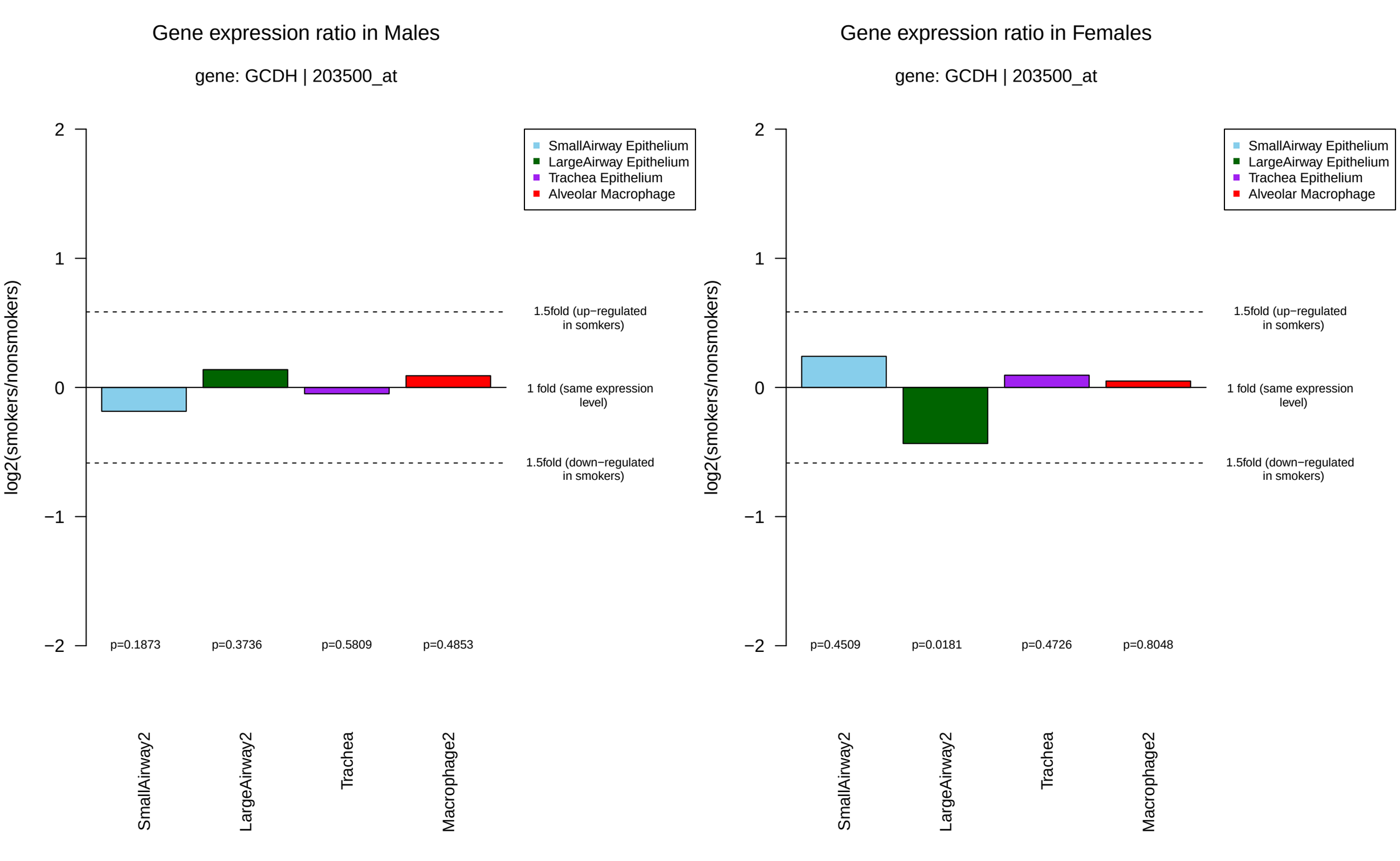
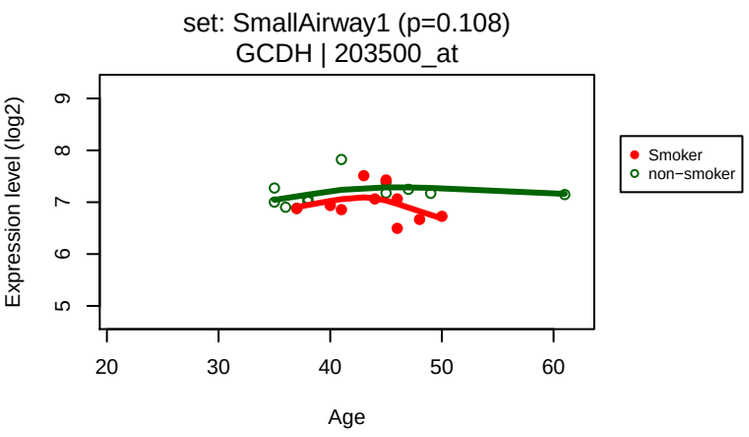
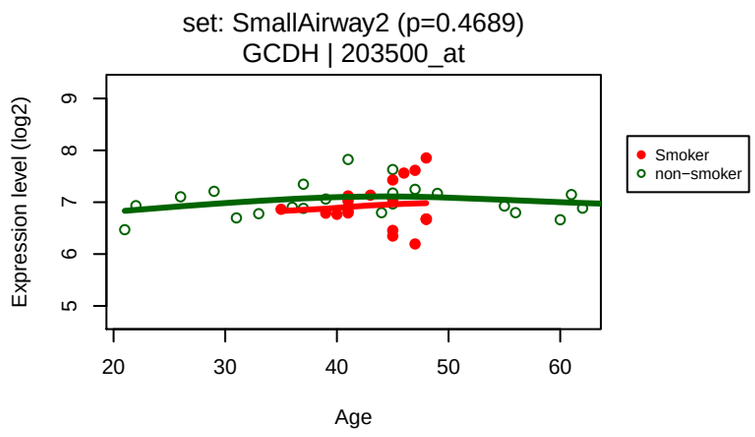
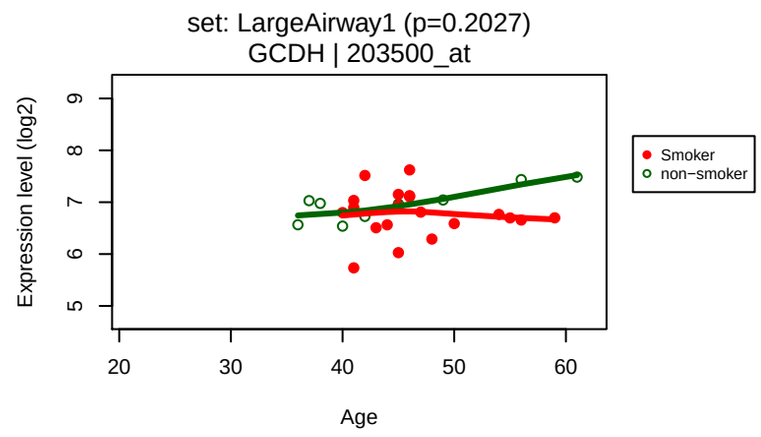

Gene expression of smokers and nonsmokers
gene: GCDH | 203500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.108
p=0.4689
p=0.0812
p=0.2027
p=0.5897
p=0.8637
p=0.7007
p=0.4158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

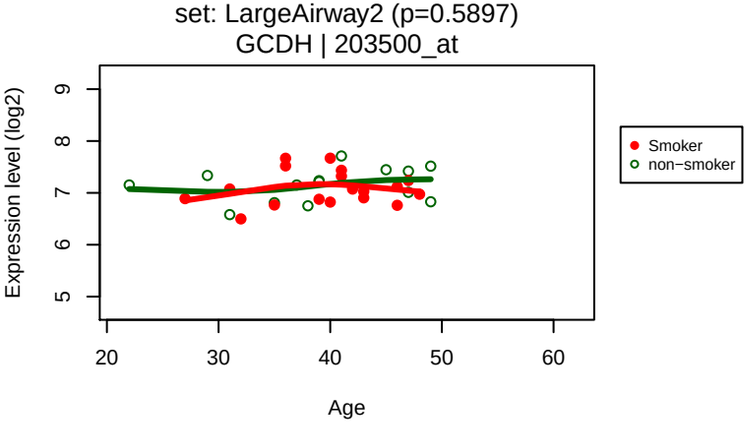

Gene expression ratio (smokers/nonsmokers)
gene: GCDH | 203500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.108
p=0.4689
p=0.0812
p=0.2027
p=0.5897
p=0.8637
p=0.7007
p=0.4158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

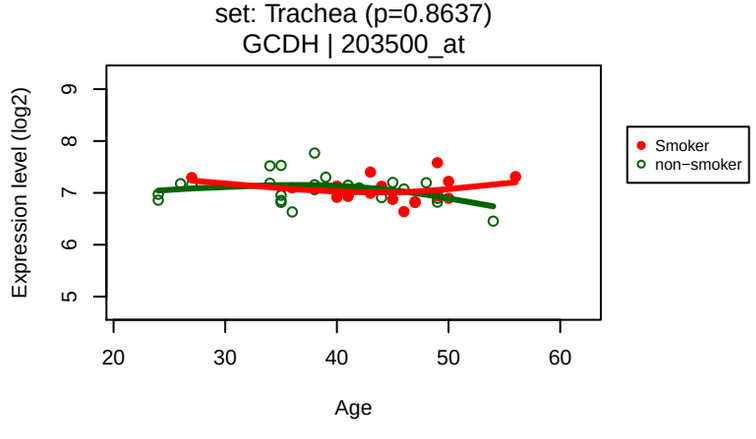

Gene expression ratio in Males
gene: GCDH | 203500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1873
p=0.3736
p=0.5809
p=0.4853
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCDH | 203500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4509
p=0.0181
p=0.4726
p=0.8048
SmallAirway2
LargeAirway2
Trachea
Macrophage2

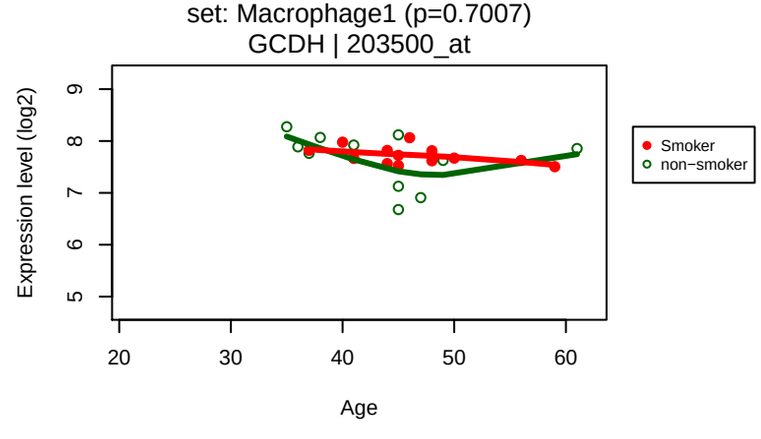

set: SmallAirway1 (p=0.108)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker

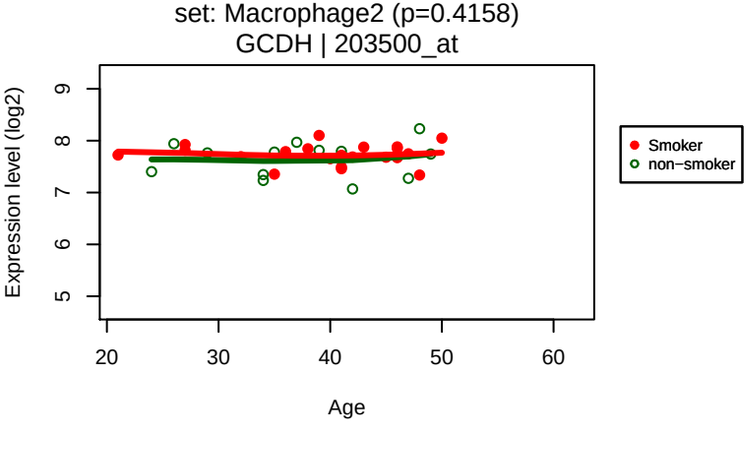

set: SmallAirway2 (p=0.4689)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker

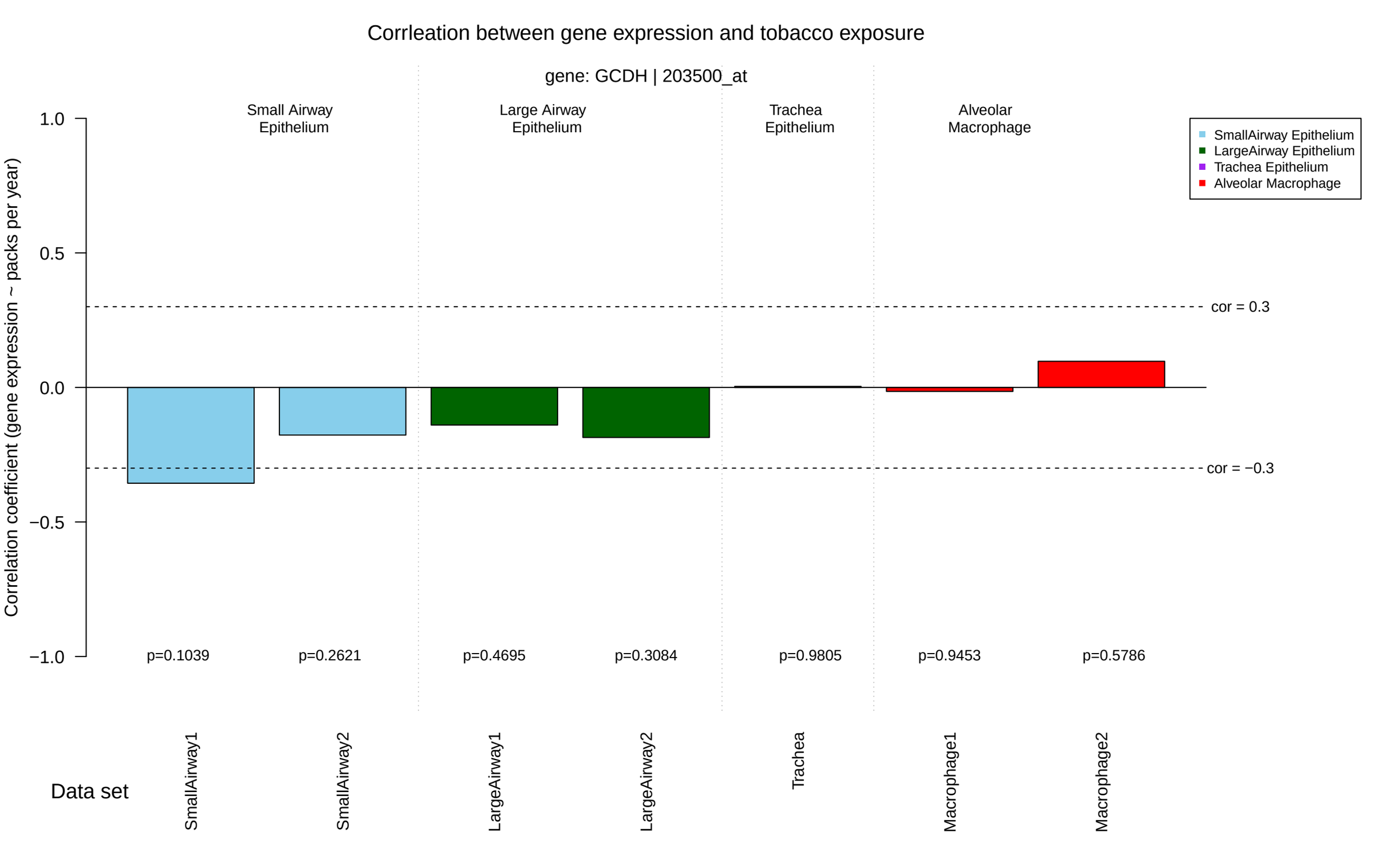

set: LargeAirway1 (p=0.2027)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5897)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8637)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7007)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4158)
GCDH | 203500_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCDH | 203500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1039
p=0.2621
p=0.4695
p=0.3084
p=0.9805
p=0.9453
p=0.5786
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2